Solution

Impulse Response Part II

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1. The impulse response of two systems connected in parallel is the _____ of the individual system impulse responses.

a) sum

b) convolution

c) integral

d) derivative

2. The impulse response of two systems connected in series, i.e., cascade, is the _____ of the individual system impulse responses.

a) sum

b) convolution

c) integral

d) derivative

3. If the absolute value of all the values in the impulse response sum to a finite number, the system guaranteed to be _____ .

a) causal

b) stable

c) memoryless

d) finite impulse response

4. Three LTI systems have impulse responses $h_1[n], h_2[n]$, and $h_3[n]$. They are connected to obtain an LTI system with impulse response $h[n] = h_1[n] * (h_2[n] + h_3[n])$. Which of the following interconnections are equivalent to $h[n]$?

a) $h_2[n] * (h_1[n] + h_3[n])$

b) $h_2[n] * h_1[n] + h_1[n] * h_3[n]$

c) $h_1[n] * h_2[n] * h_3[n]$

d) $(h_2[n] + h_3[n]) * h_1[n]$

e) $h_2[n] * h_1[n] + h_3[n] * h_1[n]$

f) $h_2[n] * h_1[n] + h_2[n] * h_3[n]$

$$h[n] = h_1[n] * (h_2[n] + h_3[n])$$

$$= h_1[n] * h_2[n] + h_1[n] * h_3[n]$$

$$= h_2[n] * h_1[n] + h_1[n] * h_3[n] \quad \text{(b)}$$

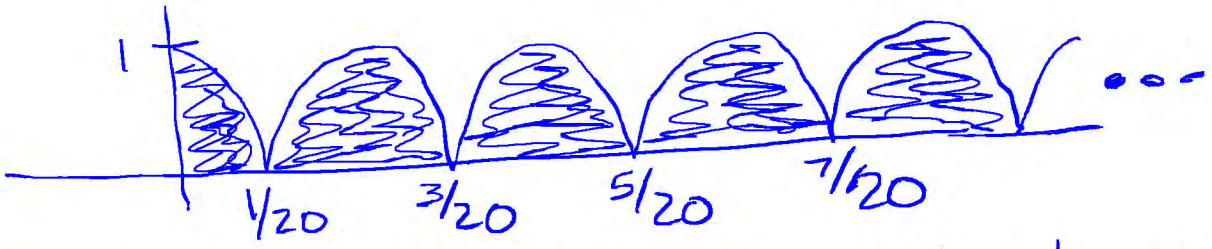
$$= h_2[n] * h_1[n] + h_3[n] * h_1[n] \quad \text{(e)}$$

$$= (h_2[n] + h_3[n]) * h_1[n] \quad \text{(d)}$$

5. A continuous-time system has impulse response $h(t) = \cos(10\pi t)u(t)$. Is this system stable?

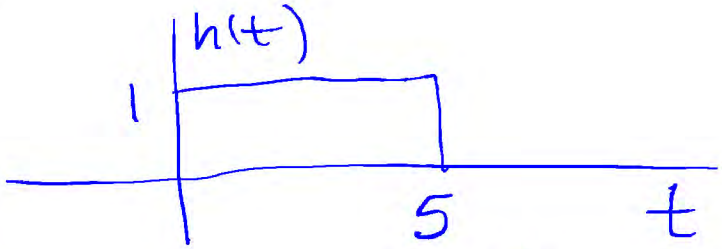
Stability requires $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$

$$\int_{-\infty}^{\infty} |h(t)|\,dt = \int_{0}^{\infty} |\cos(10\pi t)|\,dt$$



area (integral) is unbounded $\Rightarrow$ unstable

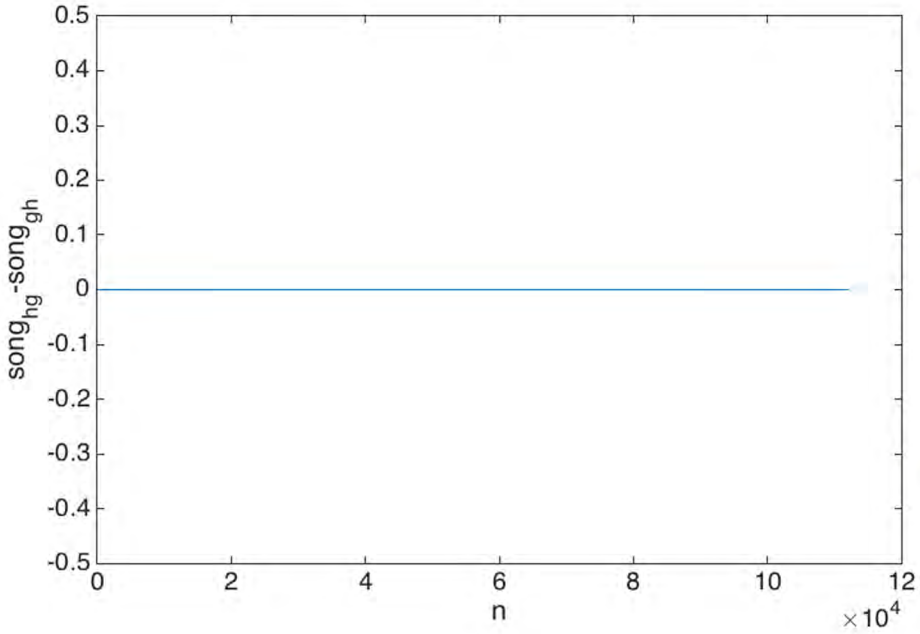
6. A continuous-time system has impulse response $h(t) = u(t) - u(t-5)$. Is this system stable?



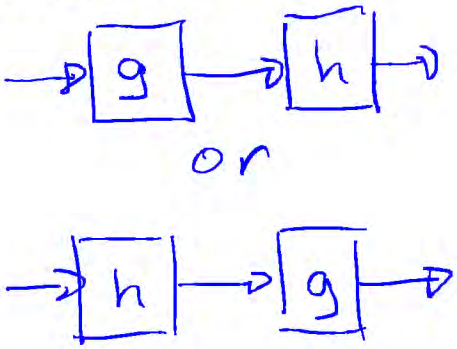
$$\int_{-\infty}^{\infty} |h(t)|\,dt = \int_{0}^{5} 1\,dt = 5$$

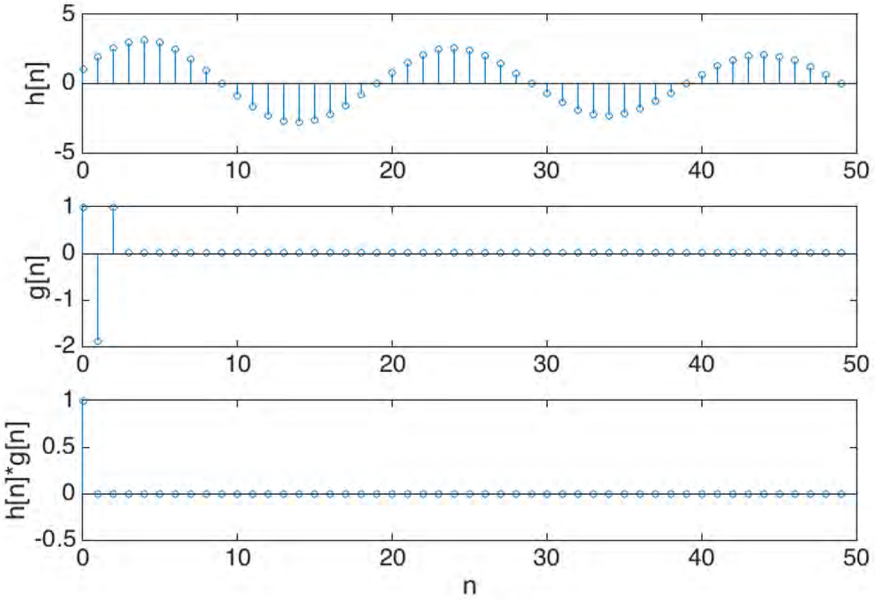
finite, so system is stable

Download `Commute.m` and `LTI.mat` from the website. `Commute.m` processes a short audio clip in two ways using systems with two different impulse responses, `h` and `g`. First it convolves `song` with `h` followed by `g`. Next, it convolves `song` with `g` followed by `h`. If convolution is commutative, these two procedures should yield identical audio clips. `Commute.m` plots the first 50 values of `h`, `g`, `h*g` and the difference between `song*h* g` and `song*g*h`. It also plays the original clip `song`, `song*h`, `song*g`, `song*h*g`, and `song*g*h`. If convolution is commutative, then the difference between `song*h*g` and `song*g*h` should be zero (or numerically very small) and `song*h*g` and `song*g*h` should sound the same.



identical output whether





7. What signal is the best description for $h[n] * g[n]$

a) $u[n]$

b) $(0.9)^n u[n]$

c) $\sin(\pi n) u[n]$

d) $\delta[n]$

e) $\cos(\pi n) u[n]$

the cascade of these particular systems results in a system whose impulse response is an impulse, that is,

$\delta[n] \longrightarrow \delta[n]$

8. Based on the answer to the previous question, the cascade of $h[n]$ and $g[n]$ _____ .

a) attenuates low frequencies

b) enhances low frequencies

c) leaves the signal unchanged

d) attenuates high frequencies

e) enhances high frequencies

an LTI system that produces an impulse in response to an impulse has output identical to its input!